

Comparative Analysis of Features, Benefits, and Gaps in Academic Journal Websites

Ar-jay F. Llanera
College of Information Technology
Central Bicol State University of Agriculture - Sipcot
Sipocot, Philippines
ar-jay.llanera@cbsua.edu.ph

Griezel An Manalo
College of Information Technology
Central Bicol State University of Agriculture - Sipcot
Sipocot, Philippines
griezelan.manalo@cbsua.edu.ph

Jeus R. Ofenda
College of Information Technology
Central Bicol State University of Agriculture - Sipcot
Sipocot, Philippines
jeus.ofenda@cbsua.edu.ph

Kenneth B. Marquez
College of Information Technology
Central Bicol State University of Agriculture - Sipcot
Sipocot, Philippines
kenneth.marquez@cbsua.edu.ph

Abstract—The online environment of an academic journal website provides researchers and students with extensive access to scholarly materials. Several available platforms, such as Scopus, Web of Science, PubMed, ERIC, IEEE Xplore, ScienceDirect, the Directory of Open Access Journals (DOAJ), and JSTOR, have distinct natures that meet specific types of user needs. Against this background, this piece of research provides a comparative analysis of these web-based resources, focusing mainly on their features, strength, and weaknesses. The study's structured methodology focuses on usability, accessibility, comprehensiveness, and technological integration. Results show the strengths of each individual platform while pointing out gaps to be filled for improvement in order to better serve the academic community. In filling those gaps, the platforms may more closely meet the emerging needs of users in the academia and beyond.

Keywords—academic journal websites, user experience, accessibility, search functionality, scholarly tools.

I. INTRODUCTION

Academic journal websites are very important for the dissemination of knowledge, offering a vast number of peer-reviewed researches. These sites help to achieve academic and professional goals but vary significantly in functionality. Differences in usability, accessibility, and content availability influence their adoption and effectiveness among users. Understanding these differences will optimize their use in the workflow of research and add value to a global audience. This study covers eight major platforms: Scopus, Web of Science, PubMed, ERIC, IEEE Xplore, ScienceDirect, DOAJ, and JSTOR. The offerings of each platform are reviewed to identify their respective strengths, limitations, and areas for potential enhancement.

Journal platforms represent a central tool to democratize access to knowledge all around the world, and even more so to underrepresented regions and developing regions of the world. At the same time as the trend in academia toward increased inter-disciplinary thinking and more technology-led operations, the effectiveness of journal platforms defines the advancement of science and scholarship. Nevertheless, systematic comparative studies of such platforms are rare and lack attention when discussing different metrics—from access to usability and even more so technology innovation.

It will emphasize isolated features, such as user experience or technical capabilities but will not give an overall review. Additionally, such shortcomings as cost barriers and limited multilingual support along with the underrepresentation of the emerging fields are not well handled. This research attempts to fill in these gaps in providing a structured multidimensional analysis that informs both the academics, institutions, and developers on how best to optimize these platforms towards accessing scholarly resources more widely and equitably.

II. RELATED WORK

Features and User Experience (UX) of Journal Websites

Academic journal websites have been increasingly designed with enhanced features to improve user experience (UX). Jackson et al. (2017) discuss the integration of article search engines, citation tools, and personalized recommendation systems to foster an intuitive, efficient navigation system, which greatly enhances usability for academic researchers [1]. Smith et al. (2018) emphasize the importance of accessibility, examining how these platforms can better accommodate diverse user groups, particularly with regard to inclusive design practices that allow for easier navigation and reading by those with disabilities [2]. Moreover, user-centered design has been central in UX research, with several studies focusing on streamlining features to support researchers' needs and boost engagement with content.

Benefits of Digital Journal Platforms

Digital journal platforms have revolutionized scholarly communication, offering numerous benefits. Williams (2019) explored how open-access models on journal websites have greatly expanded the global reach of research, allowing scholars from developing countries to access high-quality academic resources without financial barriers [3]. Similarly, Chen and Zhou (2020) discuss how the transformation of journal websites enables rapid publication and peer review, fostering interactivity through comment sections, which enhances collaboration among researchers [4]. Turner and Thomas (2016) underscore the role of advanced search capabilities and interactive data visualization tools, which streamline the academic research process by improving data accessibility and clarity for users [5].

Gaps in Journal Website Functionality

While digital journal platforms have advanced, significant gaps still exist in their functionality. Harris and Liu (2019) highlight challenges related to content discoverability and metadata quality, particularly in lesser-known journals, which can limit a researcher's ability to access and cite relevant studies [6]. Meyer et al. (2021) address the barrier for non-native English speakers, as many academic journal websites lack multilingual support, further hindering access for a global academic community [7]. Parker et al. (2018) point out the lack of integration between journal platforms and alternative scholarly communication tools, such as preprint servers or data-sharing repositories, limiting the broader dissemination and visibility of research [8].

Comparative Studies of Academic Journal Websites

a) Comparative studies have also been conducted to evaluate the differences between commercial and open-access journal platforms. Gonzalez et al. (2020) compared the design and user experience of leading publishers with open-access journals, revealing that open-access platforms prioritize user engagement through interactive features such as social media

integration and article commenting [9]. O'Brien and Kessel (2022) suggest that the incorporation of artificial intelligence (AI) tools, such as automated citation analysis and manuscript recommendations, can streamline the research process and improve user experience [10]. However, Patel and Wray (2018) highlight the challenges in the adoption of these technologies across all platforms, leading to uneven user experiences across publishers [11].

Impact of Mobile and Emerging Technologies

As mobile device use becomes ubiquitous in academic research, optimizing journal websites for mobile has become a priority. Williams and Zhang (2021) discuss how mobile optimization improves the accessibility of journal websites on smartphones and tablets. However, issues remain with features such as PDF viewing and large data set downloads, which are not yet fully optimized for mobile devices [12]. Emerging technologies such as virtual reality (VR) and augmented reality (AR) are also influencing the design of academic journal websites. Johnson et al. (2023) suggest that these technologies could revolutionize the presentation of academic articles, making content more immersive through interactive data visualizations and virtual simulations [13].

Sustainability and Environmental Considerations

The environmental impact of digital publishing has recently gained attention. Blake et al. (2023) explored how the increasing size of multimedia content, including high-resolution images and videos, contributes to the energy consumption of academic journal websites. They suggest that publishers adopt more sustainable practices, such as compressing files and optimizing digital content, to reduce the environmental footprint of these platforms [14].

III. METHODOLOGY

This study, therefore, employs a structured methodology combining qualitative and quantitative approaches for the evaluation of the platforms. This approach ensures comprehensive and objective comparison of the academic journal platforms based on predefined criteria.

Selection Criteria

These eight platforms: Scopus, Web of Science, PubMed, ERIC, IEEE Xplore, ScienceDirect, DOAJ, and JSTOR, were selected for their global relevance, their multidisciplinary scope, and features of accessibility. These are subscription-based as well as open-access sites that suit the diversified requirements of the user base. The citation-tracking features are quite advanced on Scopus and Web of Science, which are used in large parts of academia. PubMed and ERIC have been selected due to the easy accessibility and discipline-specific coverage: biomedical sciences for PubMed and education for ERIC.

IEEE Xplore and ScienceDirect were chosen because of the vast technical and scientific content, while DOAJ was included as an advocate for open-access publishing. JSTOR was selected because of its rich archival content in humanities and social sciences. The selection has ensured a balanced representation of disciplines, access models, and technological features.

Scoring System

A comparative benchmarking approach has been used to evaluate the various dimensions among which the platforms are compared. Ratings were given from a scale of 1-5 according to the criteria as follows:

Features: Functionality such as search capabilities, citation management tools, and platform integrations.

Usability: Interface design, navigation ease, and customization options.

Accessibility: Availability of open access, multilingual support, and mobile optimization.

Content Scope: Breadth, depth, and currency of the available content.

The scoring system was transparent and consistent. Public documentation, platform-specific reports, and usability studies on every dimension served as the foundation for the ratings. Reducing bias through average ratings from multiple evaluators,.

Evaluation Process

Feature Analysis: Documentation and comparison of functionalities using platform documentation and user guides.

Heuristic Evaluation: A usability check according to established design principles and usability heuristics.

Content Review: Sampling and qualitative reviews to ascertain the scope and relevance of content.

Gap Analysis: Determine functional, access as well as interdisciplinarity weakness. Comparative Benchmarking: Adding up scores across dimensions to identify relative strength and weakness. This methodology ensures that the evaluation is stringent and balanced, bringing actionable insights to stakeholders who want to optimize the use of these platforms.

IV. RESULTS AND ANALYSIS

The comparative analysis is summarized across multiple dimensions in the following table

TABLE I. ACADEMIC JOURNALS PLATFORM EVALUATION

Platform	Features (5)	Usability (5)	Accessibility (5)	Content Scope (5)	Benefits	Gaps
Scopus	5	4	3	5	Robust analytics, excellent citation tracking	High subscription cost, limited full-text access
Web of Science	5	3	3	5	Detailed citation metrics, cross-disciplinary insights	Complex interface, expensive subscription
PubMed	4	4	5	4	Free access to biomedical research, user-friendly	Narrow disciplinary focus
ERIC	3	3	5	3	Free access to education resources	Outdated interface, limited search tools
IEEE Xplore	4	4	3	5	Comprehensive technical content, advanced search tools	High subscription fees, narrow scope outside technology
ScienceDirect	5	4	3	5	Extensive scientific database, seamless integrations	High subscription fees, limited open-access content
DOAJ	3	4	5	4	Free access to open-access journals, promotes equity	Variable quality control, limited metrics
JSTOR	4	4	3	4	Archival content in humanities and social sciences	Limited recent publications, subscription costs

The table provides a comparative overview of eight major academic databases, evaluating their features, usability, accessibility, content scope, benefits, and gaps. While databases like Scopus and Web of Science excel in offering robust citation tracking and cross-disciplinary content, they face challenges in terms of usability and high subscription costs. Platforms such as PubMed and DOAJ stand out for their accessibility and free resources, though PubMed is limited to biomedical research, and DOAJ suffers from variable quality control. ScienceDirect and IEEE Xplore provide extensive scientific content and advanced tools, but also contend with high subscription fees and limited open access. Overall, the databases offer distinct advantages based on discipline and user needs but are often hindered by issues related to cost, usability, and content limitations.

TABLE II. CONTENT AND ACCESSIBILITY FEATURES

Platform	Open Access Availability	Indexed Journals	Multidisciplinary Coverage	Free Features	Subscription Requirements
Scopus	Partial	25,000+	Yes	Abstracts, search filters	Required for full access
Web of Science	Partial	21,000+	Yes	Citation metrics	Required for full access
PubMed	Full	30,000+	No	Full-text (select journals)	None
ERIC	Full	1,300+	No	Full-text Abstracts	None
IEEE Xplore	Partial	5,000+	No	Abstracts	Required for most content
ScienceDirect	Partial	2,500+	Yes	Abstracts	Required for most content
DOAJ	Full	18,000+	Yes	Full-text, free-to-use	None
JSTOR	Partial	12,000+	Yes	Some open-access content	Required for most content

The table outlines the content and accessibility features of eight academic databases, focusing on their open access availability, indexed journals, coverage, free features, and subscription requirements. PubMed and ERIC offer full access to content without requiring subscriptions, with PubMed providing free full-text access to a wide array of biomedical journals and ERIC offering open access to education-related content. In contrast, DOAJ provides full open access across a broad range of multidisciplinary journals, making it particularly attractive for free use. Other platforms like Scopus, Web of Science, IEEE Xplore, and ScienceDirect provide partial open access and require subscriptions for full access to their content. These platforms often offer valuable free features like abstracts and citation metrics but are limited in their free content. JSTOR also operates on a partial open access model, offering some free content but requiring a subscription for most resources. This highlights the trade-off between free and premium content access, with subscription-based platforms generally providing more comprehensive resources and tools.

TABLE III. TECHNICAL FEATURES AND USABILITY TOOLS

Platform	Advanced Search Filters	Citation Management Integration	Mobile Compatibility	Analytics and Metrics
Scopus	Yes	Yes	Yes	Yes
Web of Science	Yes	Yes	Yes	Yes
PubMed	Yes	Limited	Yes	Limited
ERIC	Yes	No	Yes	No

IEEE Xplore	Yes	Yes	Yes	Yes
ScienceDirect	Yes	Yes	Yes	Yes
DOAJ	Yes	No	Yes	No
JSTOR	Yes	Yes	Yes	Limited

Table III compares the technical features and usability tools across eight academic platforms. Scopus, Web of Science, IEEE Xplore, and ScienceDirect stand out for offering comprehensive usability features, including advanced search filters, citation management integration, mobile compatibility, and analytics/metrics. These platforms are designed to enhance the user experience by facilitating efficient research and citation tracking. In contrast, PubMed and DOAJ offer limited citation management and analytics, though they maintain mobile compatibility and advanced search filters. ERIC and DOAJ lack citation management and analytics tools, which may limit their effectiveness for researchers requiring in-depth tracking and analysis. Overall, the platforms with the most robust technical tools, such as Scopus and ScienceDirect, offer the most seamless and integrated user experiences for researchers.

V. DISCUSSION

The comparative analysis highlights the differences in the strengths and weaknesses of the academic journal platforms. Scopus and Web of Science have a better citation tracking and analytics advantage but are weak on cost-related barriers and lack full texts. PubMed is highly accessible for biomedical sciences but lacks cross-disciplinary scope. Similarly, ERIC has valuable content focusing on education but usability is an issue. The best quality of technical and scientific content is on IEEE Xplore and ScienceDirect, but most users cannot afford it financially. DOAJ is a great champion for open-access principles, but at best, the quality control is poor. JSTOR is an excellent archive but has few recent publications. Altogether, these results point to a need for increased accessibility, cost-effective models, and better interdisciplinary integration between sites..

VI. Recommendation

The identified gaps can be addressed by the following recommendations:

Improved Accessibility: Encourage broader adoption of open-access policies and partnerships with institutional libraries to mitigate subscription costs.

User-Centric Design: Invest in user-friendly interfaces with customizable features for different user types, including students, researchers, and professionals.

Enhanced Integration: Enable seamless connectivity with citation management tools, such as Zotero and EndNote, to streamline workflows.

Interdisciplinary Focus: Extend the content scope to include interdisciplinary research areas and emerging fields.

Flexible Pricing Models: Adopt freemium or tiered subscription models to adapt to different user needs and budgets.

VII. Conclusion

Academic journal platforms continue to play an essential role in the communication and dissemination of research. Every

platform has some unique strengths suited to a particular type of user group, but addressing their weaknesses can be really useful for them. Encouraging collaboration between publishers, researchers, and institutions can enable these platforms to adapt dynamically to the changing needs of the academic community and society at large. Such improvement is important to ensure fair access to knowledge and global academic development.

REFERENCES

- [1] A. Jackson et al., "Evaluating the Design and Usability of Academic Journal Websites," *Journal of Web Design and Development*, vol. 7, no. 2, pp. 15-25, 2017.
- [2] E. Smith et al., "Accessibility and Usability on Journal Platforms: A Usability Study," *International Journal of Digital Accessibility*, vol. 12, no. 1, pp. 32-45, 2018.
- [3] L. Williams, "Global Reach and Open Access: Facilitating Access for Developing Countries," *Journal of Academic Publishing*, vol. 45, no. 2, pp. 113-125, 2019.
- [4] X. Chen and Y. Zhou, "Digital Journal Platforms and Research Dissemination," *Journal of Scholarly Communication*, vol. 21, no. 4, pp. 198-210, 2020.
- [5] R. Turner and D. Thomas, "Enhancing Search and Data Visualization in Academic Publishing," *Library Technology Reports*, vol. 52, no. 6, pp. 45-56, 2016.
- [6] P. Harris and W. Liu, "Content Discoverability and Metadata Quality on Academic Journal Websites," *Digital Library Review*, vol. 25, no. 3, pp. 81-95, 2019.
- [7] A. Meyer et al., "Multilingual Support in Digital Scholarly Platforms: Barriers for Non-Native English Speakers," *Journal of Global Scholarly Communication*, vol. 32, no. 2, pp. 115-130, 2021.
- [8] T. Parker et al., "Integrating Preprints and Data Repositories with Journal Websites: A Gap in Scholarly Communication," *Information Science and Technology Journal*, vol. 39, no. 1, pp. 56-70, 2018.
- [9] J. Gonzalez et al., "Comparing Commercial and Open-Access Journal Websites," *Library Technology Reports*, vol. 56, no. 7, pp. 72-85, 2020.
- [10] P. O'Brien and C. Kessel, "Artificial Intelligence in Academic Journal Websites: Integration and Impact," *Journal of Digital Publishing Technology*, vol. 44, no. 3, pp. 56-69, 2022.
- [11] D. Patel and F. Wray, "Challenges in the Adoption of AI Tools in Journal Platforms," *Information Systems Journal*, vol. 33, no. 4, pp. 112-126, 2018.
- [12] J. Williams and R. Zhang, "Mobile Optimization of Academic Journal Websites: Challenges and Opportunities," *Journal of Mobile Web Development*, vol. 22, no. 1, pp. 34-47, 2021.
- [13] M. Johnson et al., "Immersive Technologies in Academic Publishing: The Future of VR and AR," *Journal of Emerging Technologies in Scholarly Communication*, vol. 11, pp. 56-72, 2023.
- [14] R. Blake et al., "Sustainability in Digital Publishing: Reducing the Environmental Impact of Journal Websites," *Sustainable Publishing Review*, vol. 29, pp. 18-31, 2023.

